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============================================================
Lab 6 - Basic Forecasting with Model Selection
============================================================
Loaded dataset: dataset.mat
Number of observations: 1247
Training observations:  1200
Forecast observations:  47
Forecast horizon:       1201 to 1247
Confidence level:       95%

============================================================
Linear Regression Candidate Models
============================================================
Model                                                          Params    Train
MSE          MAE
        RMSE     Med CRPS
LR1: constant + daily 24                                            3
521883.168      538.035
     791.283      260.300
LR2: linear trend + daily 24                                        4
513540.892      458.657
     710.656      232.094
LR3: linear trend + daily 24 + weekly 168                           6
498776.282      419.878
     675.712      209.130
LR4: quadratic trend + daily 24 + weekly 168                        7
497531.338      437.483
     702.799      215.583
LR5: linear trend + daily 24 + half-daily 12 + weekly 168           8
490046.575      402.893
     644.796      235.488

Best Linear Regression model by median CRPS:
  LR3: linear trend + daily 24 + weekly 168

============================================================
Filter Candidate Models
============================================================
Model                                  Order      Res Std          MAE
RMSE     Med CRPS

F1: Delta_24                              24      904.672      475.487
734.967      310.531

F2: Delta_1 Delta_24                      25     1110.336      753.587
1043.372     1338.159

F3: Delta_168                            168      955.139      616.918
864.900      296.036

F4: Delta_1 Delta_168                    169     1176.400      970.518
1219.322     1402.320

F5: Delta_24 Delta_168                   192     1267.724      832.639
1308.248      456.351


Best Filter model by median CRPS:
  F3: Delta_168

============================================================
Final Selected Models Comparison
============================================================
Selected model                               MAE         RMSE     Coverage
```

```
Med CRPS
LR3: linear trend + daily 24 + weekly 168      419.878      675.712       97.87%
209.130
F3: Delta_168                                616.918      864.900       97.87%
296.036

Overall, the selected Linear Regression model has better median CRPS.

Saved figures as PNG files in the current directory.

Saved model comparison metrics to lab6_model_comparison_results.csv.
Execution completed successfully.
```